

学術論文

査読付きジャーナル論文

1. Wenbo Dong, Dipankar Bhattacharya, Kai Tang, Akinari Kobayashi, **Fuyuki Tokuda**, Akira Seino, Norman Tien, Kazuhiro Kosuge, “Robotic Fabric Alignment System for Sewing Using Global Local Weighted ICP,”
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2. Akira Seino, Akinari Kobayashi, **Fuyuki Tokuda**, Kai Tang, Kazuhiro Kosuge, “Design and Development of an End-Effector for Fabric Edge Folding Along a Convex Curve With Multiple Pleats,”
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3. Hu Cheng, **Fuyuki Tokuda**, Kazuhiro Kosuge, “Automated Action Generation Based on Action Field for Robotic Garment Smoothing and Alignment,”
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4. **Fuyuki Tokuda**, Akira Seino, Akinari Kobayashi, Kai Tang, Kazuhiro Kosuge, “Transformer Driven Visual Servoing for Fabric Texture Matching Using Dual-Arm Manipulator,”
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5. Bingchen Jin, Akinari Kobayashi, Dipankar Bhattacharya, Akira Seino, **Fuyuki Tokuda**, Norman Tien, Kazuhiro Kosuge, “Automated Straight-Line Sewing of Stretchable Fabrics With Different Lengths,”
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7. Akinari Kobayashi, Wenbo Dong, Akira Seino, **Fuyuki Tokuda**, Kazuhiro Kosuge, “RollUP: Rolling-up End-effector for Fabric Handling,”
IEEE Robotics and Automation Letters (RA-L), vol. 10, no. 11, pp. 11204–11211, 2025.
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8. Letian Li, **Fuyuki Tokuda**, Akira Seino, Akinari Kobayashi, Norman C. Tien, Kazuhiro Kosuge, “Fabric Dynamic Motion Modeling and Collision Avoidance With Oriented Bounding Box,”
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9. Kai Tang, Xuzhao Huang, Akira Seino, **Fuyuki Tokuda**, Akinari Kobayashi, Norman C. Tien, Kazuhiro Kosuge, “Fixture-Free Automated Sewing System Using Dual-Arm Manipulator and High-Speed Fabric Edge Detection,”
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10. Xuzhao Huang, Akira Seino, **Fuyuki Tokuda**, Akinari Kobayashi, Dayuan Chen, Yasuhisa Hirata, Norman C. Tien, Kazuhiro Kosuge, “SIS: Seam-Informed Strategy for T-shirt Unfolding,”
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16. Yajun Xu, Shogo Arai, **Fuyuki Tokuda**, Kazuhiro Kosuge, “A Convolutional Neural Network for Point Cloud Instance Segmentation in Cluttered Scene Trained by Synthetic Data Without Color,”
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3. Letian Li, **Fuyuki Tokuda**, Akira Seino, Akinari Kobayashi, Norman C. Tien, Kazuhiro Kosuge, “Modeling of Dynamic Fabric Motion Using Multiple Oriented Bounding Boxes,”
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4. Edmund Lo, Xuzhao Huang, Kai Tang, Akira Seino, **Fuyuki Tokuda**, Kazuhiro Kosuge, “Fabric Flattening and Alignment System Using Real-time Mesh-based State Estimation and Visual Servoing,”
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Best Conference Paper Award

5. Wenbo Dong, Dipankar Bhattacharya, Akinari Kobayashi, Akira Seino, **Fuyuki Tokuda**, Xuzhao Huang, Kai Tang, Norman C. Tien, Kazuhiro Kosuge, "Precise Top-Layer Fabric Segmentation for Fabric Destacking with Edge- and Shape-Aware Deep Networks,"
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10. **Fuyuki Tokuda**, Akira Seino, Akinari Kobayashi, Kazuhiro Kosuge, "CNN-based Visual Servoing for Simultaneous Positioning and Flattening of Soft Fabric Parts,"
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11. Diyi Liu , Shogo Arai, **Fuyuki Tokuda**, Yajun Xu, Jun Kinugawa, Kazuhiro Kosuge, "Deep-Learning based Robust Edge Detection for Point Pair Feature-based Pose Estimation with Multiple Edge Appearance Models,"
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Best Oral Presentation Award

学術上特記すべき業績

特許

1. 「Automated sewing system and controlling method thereof」
発明者：Kazuhiro Kosuge, Kai Tang, Akira Seino, **Fuyuki Tokuda**, Akinari Kobayashi, Hoi Pang Ng
国際出願：PCT/CN2024/076693, 2024.02.07
2. 「Systems and Methods for Sewing and Un-wrinkling Fabrics」
(生地の縫製およびしわ伸ばしのためのシステムおよび方法)
発明者：小菅一弘, 徳田冬樹
国際出願：PCT/CN2022/118736, 2022.09.14
日本出願：特願 2025-513230, 2022.09.14
3. 「ロボット制御装置」
発明者：矢野巧人, 濃野真広, 小菅一弘, 荒井翔悟, 徳田冬樹
日本出願：特願 2020-003785, 2020.01.14

作品（特許展示会）

1. **Fuyuki Tokuda**, Kazuhiro Kosuge, “Smart Fixture-free 2D Sewing System with Automatic Fabric Alignment,” The 9th International Invention Innovation Competition in Canada (iCAN), Toronto, Canada, August 2024.
Gold Medal Award / Organizer's Choice Award / Inventors College Organization Award of Excellence
2. Akinari Kobayashi, **Fuyuki Tokuda**, Kazuhiro Kosuge, “Intelligent 2D Sewing System without Fixation with Automatic Fabric Alignment,” The 49th International Exhibition of Inventions Geneva, Geneva, Switzerland, April 2024.
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2. **徳田冬樹**, 荒井翔悟, 小菅一弘, 「画像ヤコビアンレスビジュアルサーボによる物体の位置決め」,
第24回ロボティクスシンポジア, 5C4, 富山, 2019年3月（口頭発表, 査読有り）

その他

受賞歴

1. IEEE International Conference on Mechatronics and Automation (ICMA) 2025, Best Conference Paper Award, August 2025.
2. The 9th International Invention Innovation Competition in Canada (iCAN), Gold Medal Award/ Organizer's Choice Award / Inventors College Organization Award of Excellence, August 2024.
3. The 49th International Exhibition of Inventions Geneva, Silver Medal Award, April 2024.
4. 計測自動制御学会東北支部 55 周年記念学術講演会, 優秀発表奨励賞受賞, 2020 年 2 月
5. International Conference on Nanoscience and Nanotechnology 2017, Best Oral Presentation Award, February 2017.

研究助成

1. 公益財団法人 電気通信普及財団, 2025 年度研究調査助成, 「生成モデルと模倣学習を融合した新たなロボット制御手法による柔軟物体操作の高精度化」, 総額 500 万円, 2026 年 4 月 - 2028 年 3 月
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講演

招待講演

1. **Fuyuki Tokuda**, “Visual Servoing Enhanced with AI,”
WRC Symposium on Advanced Robotics and Automation (WRC SARA), Beijing, China, 10 August 2025.

ワークショップ講演

1. **Fuyuki Tokuda**, “CNN-based Visual Servoing for Flattening of Fabric Parts,”
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2. **Fuyuki Tokuda**, “Fixture-Free Sewing System,”
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2. Shogo Arai, Zhuang Feng, **Fuyuki Tokuda**, Adam Purnomo, Kazuhiro Kosuge, “Deep Learning-based Fast Grasp Planning for Robotic Bin-picking by Small Data Set without GPU,”
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査読無し国内学会

1. 石井創太, 徳田冬樹, 林部充宏, 「ビジュアルフィードバックの統合による視覚言語行動モデル VLA の制御高精度化」,
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3. 徳田冬樹, 荒井翔悟, 小菅一弘, 「CNN を用いたビジュアルサーボによる把持物体の位置決め」,
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8. 徳田冬樹, 本田光裕, 奥村竜二, 市川洋「グラフェン上への配向性 ZnO ナノロッドの直接合成」,
第 63 回応用物理学会春季学術講演会, 21a-H111-6, 東京, 2016 年 3 月 (口頭発表, 査読無し)
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学位論文

1. 「Object Positioning Using Convolutional Neural Networks」
博士 (工学) 東北大学 2022 年 3 月 25 日
2. 「Image-Jacobian-Less Visual Servoing」
修士 (工学) 東北大学 2019 年 3 月 27 日
3. 「グラフェン上への ZnO ナノロッドの直接合成」
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